

Code Documentation

The project implements an Information Retrieval system for the provided 100-document clothing corpus.

1. Preprocessing and Inverted Index

- The corpus is parsed to extract document ID, category, title, and description.
- Text is converted to lowercase and tokenized.
- Punctuation is removed and words are stemmed using the Porter Stemmer.
- Stopwords are retained to preserve information needed for phrase and proximity queries.
- An inverted index is constructed containing each term's document frequency (df) and postings list with (docID, tf).

2. Ranked Free-Text Retrieval

- Document vectors are constructed using the lnc weighting scheme.
- Query vectors use the ltc weighting scheme.
- Cosine similarity is calculated between the query and document vectors.
- The system returns the top 10 documents, sorted by decreasing cosine similarity, with document ID, title, category, and score.

3. Positional Index and Phrase Search

- A positional index is constructed containing (docID, tf, positions) for each term.
- Exact phrase queries are processed using token positions.
- A document is returned only when the query terms occur consecutively and in the specified order.
- Matching ending positions are displayed in the interface.

4. Proximity Search

- Proximity queries use the format term1 WITHIN/k term2.
- The positional index is used to find occurrences where the first term appears before the second term within k token positions.
- Matching position pairs are displayed along with the corresponding documents.

5. Streamlit Interface

A Streamlit-based interface provides three search modes:

- Free Text (VSM)
- Phrase Search
- Proximity Search

The application also displays corpus information and automatically generates the inverted and positional index files.

6. Testing

The system was tested using multiple free-text, exact phrase, and proximity queries, including a query containing a term not present in the corpus. The results and representative screenshots are included in the accompanying assignment documentation.